



AI Playbook and Toolkit for Public Health Departments

Geospatial AI and Spatial Epidemiology

ANL 230

Geospatial AI combines spatial data, geographic analysis, and AI methods to identify patterns across place. In public health, spatial analysis is already central to outbreak response, environmental health, emergency preparedness, community health assessment, service planning, and health equity work. AI can strengthen these activities when it helps detect clusters, estimate access barriers, classify imagery, prioritize field work, or synthesize spatial information for decision-makers. This module introduces the technical and governance issues that arise when AI uses location. Location data can be powerful, but it can also be sensitive. Even when names are removed, detailed geography can reveal identity, community characteristics, or sensitive health patterns. Public health staff must understand geocoding quality, aggregation, spatial bias, privacy, and the limits of maps. Learners will examine how geospatial AI can support public health while avoiding common mistakes, such as treating maps as neutral, ignoring missing addresses, overinterpreting small-area patterns, or using spatial proxies that reproduce inequity.

Level	applied/foundational
Primary track	Analytics and Modeling Track
Appears in tracks	analytics-modeling

Learning Objectives

- Define geospatial AI and spatial epidemiology in public health terms.
- Explain how geocoding, aggregation, spatial scale, and data quality affect AI-supported maps and analyses.
- Identify privacy and re-identification risks related to location data.
- Recognize spatial bias, modifiable areal unit problems, and ecological interpretation risks.
- Design guardrails for AI-supported spatial analysis, hotspot detection, service planning, or emergency response.
- Produce a geospatial AI readiness worksheet for a public health workflow.

Module Overview

Geospatial AI combines spatial data, geographic analysis, and AI methods to identify patterns across place. In public health, spatial analysis is already central to outbreak response, environmental health, emergency preparedness, community health assessment, service planning, and health equity work. AI can strengthen these activities when it helps detect clusters, estimate access barriers, classify imagery, prioritize field work, or synthesize spatial information for decision-makers.

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Why This Topic Matters for Public Health AI

Geospatial AI matters because public health decisions are often place-based. Agencies decide where to conduct outreach, deploy mobile services, inspect facilities, respond to environmental hazards, monitor outbreaks, and communicate risks. AI-supported spatial tools can help analyze large volumes of geographic information and identify patterns that may not be visible in simple tables.

However, maps can create a false sense of objectivity. A hotspot may reflect where data are collected, who has access to testing, which providers report electronically, or which addresses are easier to geocode. Areas with missing or poor-quality data may appear low risk when they are actually under-observed. Spatial AI therefore requires careful review of data completeness, spatial scale, and community context.

Location data also raises privacy and trust concerns. Public health agencies have authority to collect and use sensitive information, but that authority should be matched with safeguards. When AI tools combine health data, address data, demographic data, mobility data, environmental data, or social vulnerability indicators, agencies must consider re-identification, stigmatization, public communication, and community impact.

Public Health Example

A health department may use AI-supported geospatial analysis to identify areas where mobile vaccination clinics could improve access. The tool may combine immunization coverage, school locations, transportation routes, census variables, provider locations, and appointment availability. The output may recommend priority areas for outreach and clinic placement.

This workflow can advance equity if it identifies access barriers and supports community partnership. It can cause harm if it labels neighborhoods as low-performing without context, uses outdated population denominators, ignores language access, or fails to engage local organizations. The geospatial analysis should be paired with community review, accessibility planning, and transparent explanation of limitations.

Technical Deep Dive

Geospatial AI begins with location data, and location data are rarely perfect. Addresses may be missing, misspelled, outdated, rural route-based, temporary, or unstable. People experiencing homelessness, migration, displacement, incarceration, or housing instability may be poorly represented by address-based analysis. Geocoding errors can shift records across boundaries and distort rates for small areas.

Spatial scale shapes interpretation. County-level maps may hide neighborhood-level inequities. Census tract maps may reveal meaningful local variation but increase privacy risks. ZIP codes are convenient but were created for mail delivery, not public health analysis. Staff should choose geographic units based on the question, data quality, privacy requirements, and actionability.

AI-supported hotspot detection requires baseline context. An apparent cluster may reflect higher testing volume, better reporting, a facility-specific outbreak, or a true increase in incidence. Algorithms can detect unusual patterns, but epidemiologists must assess plausibility, data quality, denominator stability, and reporting artifacts. Automated hotspot alerts should be treated as signals for review, not conclusions.

Geospatial models may use spatial proxies for social conditions. Social vulnerability, deprivation, rurality, housing burden, transportation access, and environmental exposure indicators can improve public health planning. They can also be misused if they stigmatize communities or support restrictive decisions. Agencies should use these indicators to guide supportive action and resource allocation, not blame.

Privacy protection should be planned before analysis. Techniques include aggregation, suppression, geomasking, minimum cell-size rules, role-based access, public/private map separation, and review of small-area outputs before publication. For internal AI tools, access controls and audit logs should apply to spatial data just as they do to clinical or case data.

Risks, Failure Modes, and Guardrails

Geocoding bias. Some populations and locations are harder to geocode accurately. Guardrails include geocoding quality checks, manual review for unmatched records, and documentation of missingness.

Small-area privacy risk. Detailed maps can reveal sensitive patterns or identities. Guardrails include aggregation, suppression, role-based access, and public release review.

Overinterpretation of hotspots. Spatial clusters may reflect testing or reporting patterns. Guardrails include denominator review, baseline comparison, and epidemiologic confirmation.

Stigmatizing communities. Maps can unintentionally label communities as problems. Guardrails include equity framing, community engagement, careful language, and focus on structural supports.

Application to AI-Supported Workflows

Geospatial AI may support predictive analytics, emergency response, inspection prioritization, environmental monitoring, and service planning. Generative AI may summarize spatial patterns for leaders, but it must not overstate certainty or ignore limitations. RAG tools may retrieve local plans, maps, and protocols, while agentic workflows may propose outreach routes or inspection assignments that require human approval.

The responsible application of geospatial AI requires explicit attention to privacy, geography, and equity. A spatial output should answer a public health question, support a feasible action, and include enough context for staff and

communities to interpret it appropriately.

Reflection Questions

What location data are used, and how complete and accurate are they?

Which geographic unit best matches the public health action?

Could missing or poor-quality location data affect certain populations more than others?

What privacy safeguards are needed for internal and public maps?

How will community context be included before action is taken?

Practical Exercise

Create a geospatial AI readiness worksheet for one public health workflow. Include the public health question, location data sources, geocoding method, geographic unit, denominator, privacy safeguards, equity concerns, AI-supported task, human review process, and communication plan.

The worksheet should identify at least one way the map could be misinterpreted and one safeguard that would reduce the risk of harm.

Expected Artifact or Evidence

Geospatial AI readiness worksheet

Map interpretation and limitations memo

Privacy and suppression review notes

Community context and action plan

Expected Assignment

- Geospatial AI readiness worksheet
- Map interpretation and limitations memo
- Privacy and suppression review notes
- Community context and action plan

Knowledge Check

- 1. What is geocoding?
- 2. Why can maps create misleading impressions?
- 3. What is the modifiable areal unit problem?
- 4. Which safeguard is appropriate before publishing a small-area health map?
- 5. What is strong completion evidence for this module?

References and Resources

- CDC GIS and Public Health Resources: <https://www.cdc.gov/gis/>
- NIST Artificial Intelligence Risk Management Framework: <https://www.nist.gov/itl/ai-risk-management-framework>

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- NIST Privacy Framework: <https://www.nist.gov/privacy-framework>
 - CDC Public Health Data Strategy: <https://www.cdc.gov/public-health-data-strategy/php/about/index.html>